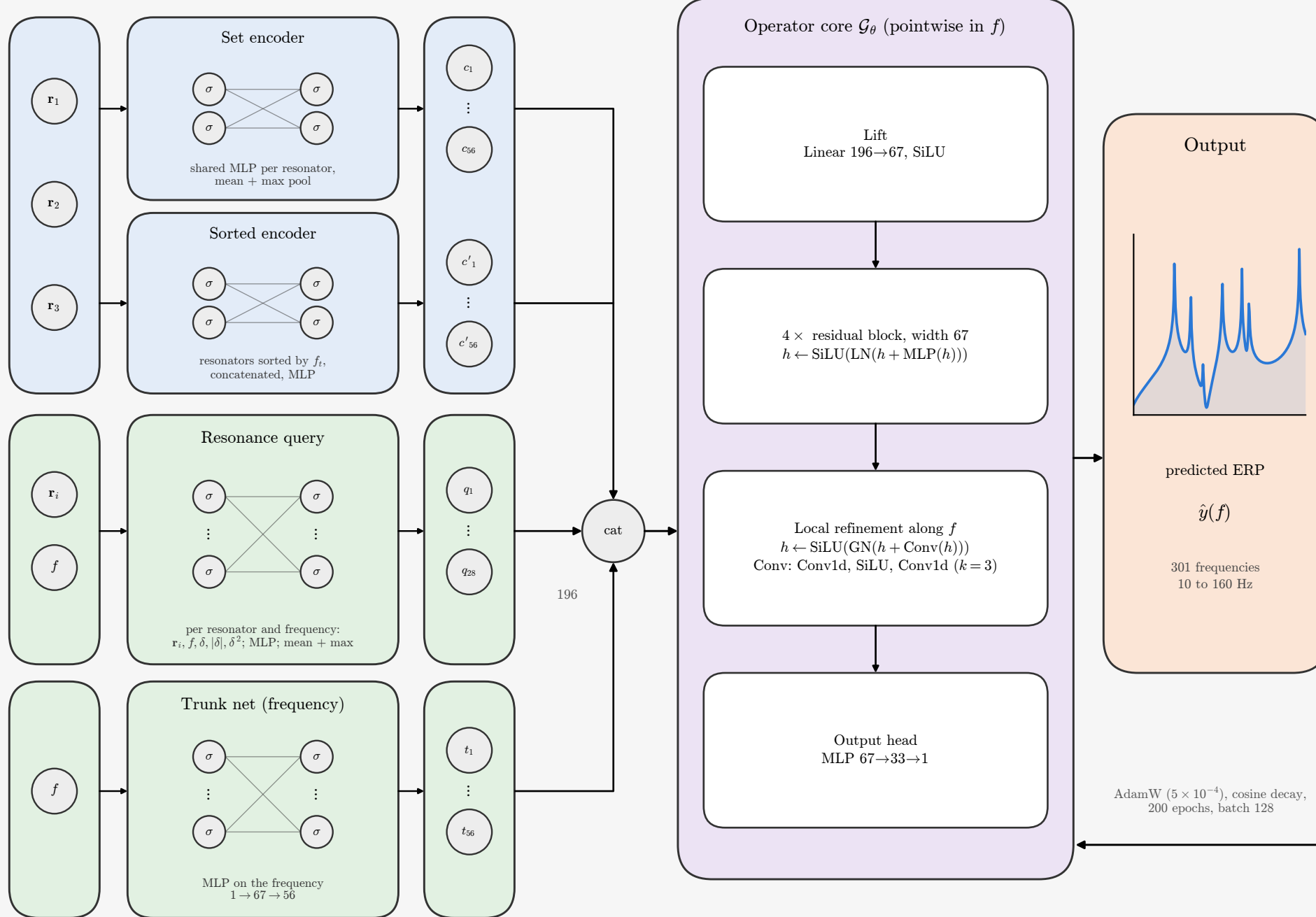


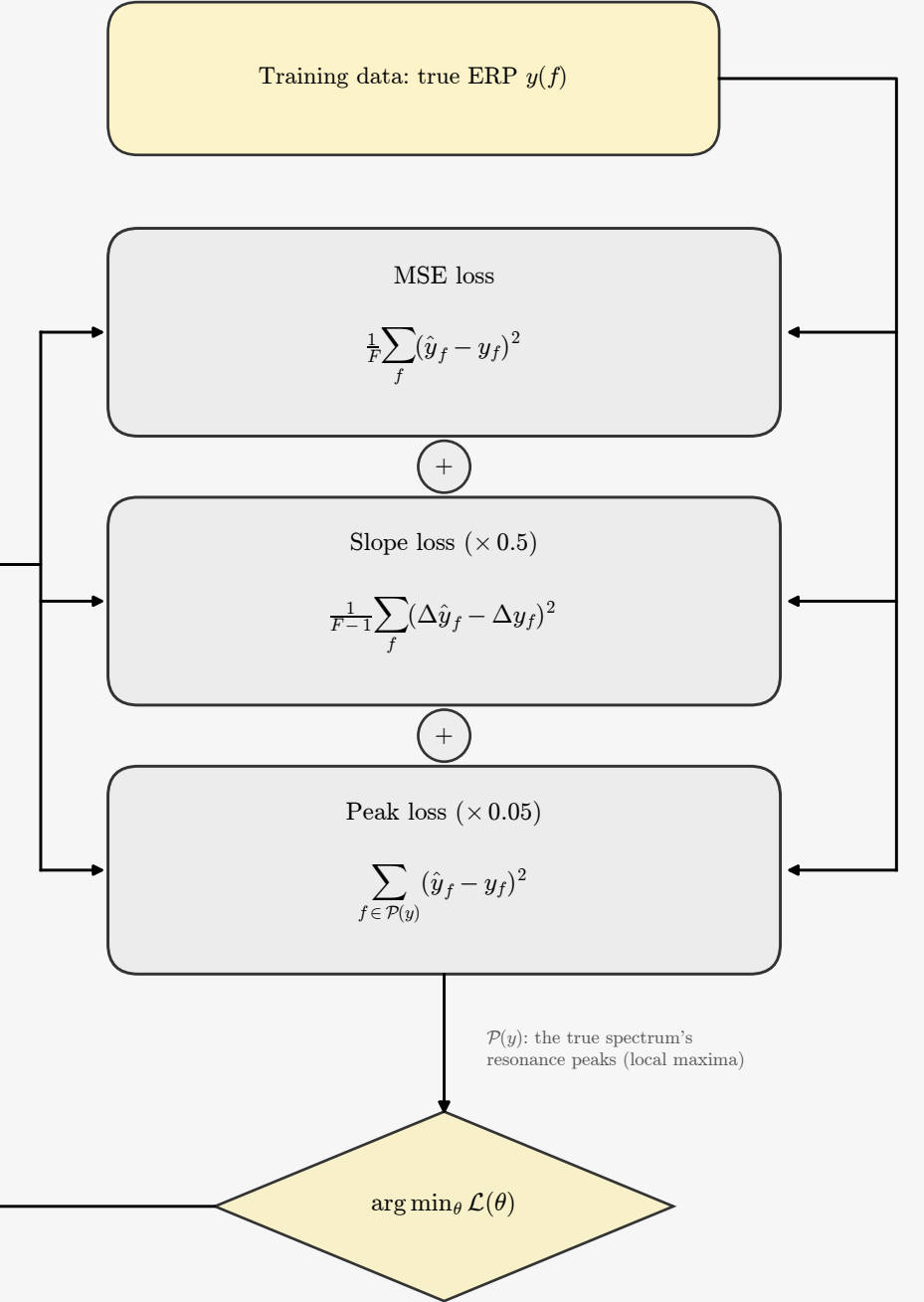
Forward neural operator: Deep Cat Operator (DCO)

resonator design and frequency $\rightarrow$ ERP spectrum; 155,240 parameters

$\mathbf{r}_i = (m_i, k_i, f_{t,i}, x_i, y_i)$, $i = 1, 2, 3$ (z-scored)



Loss functions



Each $\mathbf{r}_i$ is extended by 120 plate-mode features: $\sin(a\pi x/L_x)$, $\sin(b\pi y/L_y)$ and their products, $a, b = 1, \dots, 10$. ERP values are normalised before the loss.

Same data, output and loss for all forward operators (DON, DNO, FNO, GNO, STO, SIREN, WNO, LNO, NN); they differ in the operator core and in how the resonators are encoded. Data: 100k configurations, 3 resonators, 80/10/10 split.